

Case Studies in the Effects of Anthropogenic Change on Animal Social Behavior

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From the gradual progression of climate change to the immediate disturbances of noise and light pollution, anthropogenic change has significant implications for wildlife social behavior on multiple time scales. While the gradual force of climate change drives long-term evolutionary shifts in social systems, it also elicits short-term behavioral adjustments in response to altered environmental conditions. Conversely, the immediate disturbances of artificial light and noise often trigger rapid behavioral changes, which, if persistent over generations, may in turn contribute to the evolutionary trajectory of social behavior. The following analysis examines how gradual and immediate anthropogenic changes influence both the short-term presentation and long-term evolution of social behavior across a variety of non-human animal species.

I. Gradual Anthropogenic Change: Climate Change

a. Effects on Short-Term Behavioral Alterations

While climate change often produces evolutionary effects that unfold over extended periods of time, it is important to acknowledge the climate-driven alterations in social behavior that occur on shorter time scales as well. To combat effects of climate change like shifts in precipitation patterns, increased habitat fragmentation, or the consequences of rising temperatures, individuals often make immediate, daily decisions regarding their social behavior and interactions. For example, extreme heat conditions have been observed to influence the social network structures of domestic sheep, with individuals forming smaller, more fragmented groups with fewer repeated interactions during warm, dry periods or when shade is limited

(Borthwick et al., 2024). This short-term network alteration exemplifies immediate behavioral adjustment to a long-term, climate-driven shift in habitat conditions.

Climate change can also cause immediate behavioral changes through sensory disruption. Ocean acidification, for example, has been shown to impair the olfactory and auditory senses of certain fishes, reducing their critical ability to detect predators, kin, and safe reef locations. Clownfish raised in simulated CO₂ concentrations exhibit impaired hearing and reduced avoidance of predator-associated reef noise (Simpson et al., 2011), demonstrating how ocean acidification can disrupt the sensory cues that normally facilitate critical aspects of sociality like group cohesion, parental care, and predator avoidance. Similarly, climate-driven shifts in landscape features like snow cover and vegetation can reduce the effectiveness of camouflaging abilities, increasing visibility and ultimately predation risk in both aquatic and terrestrial species.

Also posing immediate implications on social behavior are the evolutionary dynamics of climate change and disease. As warmer, wetter conditions broaden the physical ranges and breeding periods of disease vectors like mosquitoes and ticks, individuals may be prompted to adjust their social behavior in response to this increasing threat of disease. Some may endeavor to form smaller and/or more transient groups to limit disease transmission, like certain bat colonies that, when faced with an outbreak of white-nose syndrome, reduced clustering density despite thermoregulatory costs (Langwing et al., 2012). Other individuals may remain in their typical group sizes but take actions that help manage the risk of infection, such as mandrills that avoid grooming conspecifics who display signs of parasitic infection, like worms in feces or on the skin (Poirotte et al., 2017). Each of these examples demonstrates how climate-driven disease pressures can elicit quick and temporary combative responses in individuals.

b. Effects on the Long-Term Evolution of Social Behavior

As climate change exerts evolutionary pressure on ecosystems and subsequently populations, the immediate behavioral shifts mentioned above may develop into targets of selection over time. For example, dividing into smaller groups may be a short-term avoidance reaction to the presence of parasites, but as climate change amplifies disease pressures, evolution may come to favor smaller, more modular social structures to begin with. A comparative study provides evidence for this in its finding that across 19 wild primate species, those with greater social network modularity tended to exhibit lower socially-transmitted parasite richness, ultimately suggesting that disease pressures may have favored the evolution and maintenance of more modular social structures in those species over time (Griffin & Nunn, 2011).

Disease pressures can also shape the evolutionary development and fine-tuning of social immunity strategies on a group level. While mandrills exhibit short-term avoidance of parasites through selective grooming, there are species — namely, eusocial insects — that exhibit heritable, species-wide organizational traits that operate on the colony level and that exist solely to reduce colony infection and thus increase fitness. Among these hardwired behaviors are hygienic brood removal, antimicrobial secretions for nest sanitation, isolation of infected individuals, and grooming of foragers to prevent pathogen entrance into the colony, to name just a few (Cremer et al., 2018). While the evolution of social immunity in these insects can be largely attributed to the chronic pathogen pressures intrinsic to dense group living, it does provide keen insight into the mechanisms of social immunity and how they may evolve similarly or perhaps contrastingly in other taxa as a response to climate-driven disease pressures in particular.

Like disease pressures, ocean acidification and its attendant disruption of marine social systems may lead to evolutionary shifts in the behavior of affected species. As high CO₂ concentrations reduce the reliability of social cues like kin recognition and group movement detection, maintaining tight shoals may become costly. Heritable tolerance in behavior under high CO₂ conditions, however, has been demonstrated in reef damselfish, with some families retaining typical social behavior while others lose cohesion (Welch et al., 2017). Given the reduced reliability of social cues in high CO₂ conditions and the demonstrated genetic variation in behavioral tolerance of these conditions, selection could take a couple of routes; it could favor the behaviorally tolerant genotypes that preserve social cohesion despite sensory disruption, or, if these genotypes are rare or costly, it could favor individuals that rely less on social coordination, ultimately shaping the evolution of looser, less synchronized groups in these species. Although speculative, these possible trajectories illustrate the climate-driven evolutionary shifts in social features like group structure and cohesion that may very likely occur for certain marine species in an increasingly acidic ocean.

II. Immediate Anthropogenic Change: Noise and Light Pollution

a. Effects on Short-Term Behavioral Alterations

Short-term anthropogenic disturbances often produce rapid and reversible changes in social behavior, with variables like noise and light pollution disrupting sensory processing and social interactions within days or even hours of exposure. For example, increased cliffside illumination by incoming ship lights has been observed to coincide with a reduction in colony attendance in a species of nocturnal shearwater seabird (Austad et al., 2023). A similar study on

another shearwater species reports that as cliffside illumination increases, average chick mass decreases (Cianchetti-Benedetti et al., 2018). Based on the complimentary findings of these two studies, it can be deduced that anthropogenic light pollution correlates with a decreased frequency in parent-offspring feeding events, ultimately signifying an immediate, anthropogenic disruption in the parental care behavior of shearwater birds.

In a similar manner, noise pollution is emerging as a disruptive force in the social communication of many species. In one study conducted adjacent to coastal shipping lanes, bottlenose dolphins exhibited altered whistling behaviors, counteracting the auditory masking effects of vessel activity not only by whistling at higher frequencies, but also by simplifying the contour shape (i.e. the complexity) of their calls (Fouda et. al., 2018). It is reasonable to suspect that such a sudden-onset simplification of their millenia-evolved calls may elicit a rapid breakdown in the effectiveness of bottlenose dolphins' communication strategies. This effect is certainly even harsher for species with more rigid call production capabilities, like the European tree frog which showed no calling plasticity in response to increased background noise in a particular study (Lengagne, 2008). With so many species relying heavily on vocal communication and so much variability in short-term vocal plasticity, artificial noise can immediately disrupt social cohesion and coordination, allowing more flexible individuals to adjust their signals as needed while those less adaptable are left momentarily isolated and unable to effectively communicate with conspecifics.

b. Effects on the Long-Term Evolution of Social Behavior

While noise and light pollution generally produce immediate behavioral shifts, their persistence over decades may cause evolutionary changes in sociality as well. With its immediate constraint on activity patterns, chronic light pollution may drive shearwater birds to evolve properly altered circadian rhythms or increased physiological and sensorial tolerance to light exposure. Similarly, with its immediate constraint on communication strategies, chronic noise pollution may drive bottlenose dolphins to evolve improved capacities for producing and detecting auditory signals at higher frequencies, or greater efficiency in sending and receiving the same information with simpler calls. Ultimately, the disturbances of noise and light pollution, while immediate, have the potential to shape the evolutionary course of social communication strategies in affected populations.

Conclusion

Gradual and immediate anthropogenic changes alike profoundly influence the social behavior of many species. To cope with altered environmental conditions on short time scales, individuals adjust their immediate behavior by modifying group sizes, reorganizing social networks, or changing communication strategies. Over evolutionary time, these pressures can shape the development of more modular social structures, refine social immunity mechanisms, and select for behavioral traits that maintain social cohesion or support greater independence in disrupted environments. From the above examples involving mammals, birds, fish, and insects, it

is evident that anthropogenic change not only disrupts existing social systems but also provides the selective pressures that may continue to guide the evolution of social behavior in a rapidly changing global environment.

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